When sparsity hides the actuator: a pre-registered economic benchmark of SINDy-MPC for greenhouse climate control

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Abstract

Interpretable data-driven surrogates such as sparse identification of nonlinear dynamics (SINDy) are attractive for greenhouse model predictive control (MPC) because they yield explicit, inspectable equations rather than black boxes. Whether this interpretability comes at an economic cost, however, is rarely tested under a fair protocol: baselines are often untuned, the metric is a tracking proxy rather than profit, and—most subtly—the identification recipe is chosen by open-loop prediction quality. We report a pre-registered, in-silico economic benchmark of ten controllers on the GreenLight tomato simulator (Rostov-on-Don, 2018–2023 ERA5 weather), using the simulator's own economic performance indicator (EPI, seasonal profit) as the primary metric and a two-axis Pareto criterion (EPI versus constraint violations), over 20 seeds with paired non-parametric statistics. The pre-registered SINDy-MPC recipe, frozen by open-loop metrics before any closed-loop evaluation, is economically among the weakest interpretable controllers (1.18 vs 4.63 EUR/m^2 for a tuned rule-based baseline). A one-parameter ablation traces the cause: the sparsity threshold zeroes the boiler term in the temperature equation—the only modelled heating actuator—and closed-loop EPI collapses from 3.8 to 1.0 EUR/m^2 while the open-loop rollout error stays flat (2.25 $\rightarrow$ 2.39). The open-loop selection criterion is thus insensitive to the closed-loop failure it induces. Restoring the term, by a denser threshold or by DAgger, recovers EPI to parity with the heuristic without loss of interpretability, and the repaired model coincides with a first-principles grey-box model. Comparison with additional controllers rules out the obvious alternative explanations: the true-model oracle over-spends and is not a ceiling, a normalized PPO improves but does not win, and a black-box NN-MPC is worst—so neither insufficient fidelity nor interpretability is the bottleneck. The interpretable surrogate's demonstrable value lies instead in safety and out-of-distribution guarding, not in economic supremacy. We conclude that parsimony is not interpretability, that sparsity must be removed from model-selection objectives when the downstream task is closed-loop control, and that open-loop-frozen pre-registration can systematically certify a control-catastrophic model.

Keywords: greenhouse climate control, model predictive control, sparse identification of nonlinear dynamics, interpretable machine learning, identification for control, economic benchmarking

1. Introduction

Greenhouse horticulture is an energy-intensive industry in which climate control decisions translate directly into profit: heating, CO_2 enrichment and supplemental lighting are the dominant variable costs, and they must be balanced against the revenue from crop growth. Model predictive control (MPC) is a natural fit, since it optimises a finite-horizon objective subject to the temperature, humidity and CO_2 constraints that define a productive climate [1, 2, 3]. MPC, however, needs a model, and first-principles greenhouse models are laborious to build and calibrate. This motivates data-driven surrogates, and in particular sparse identification of nonlinear dynamics (SINDy), which recovers explicit governing equations as a sparse combination of candidate terms [4, 5, 6]. Unlike neural surrogates, a SINDy model is a glass box: its equations can be read, sign-checked against physics, and embedded analytically in an MPC solver. For a grower or a certifying body, this interpretability is not a luxury but a practical requirement for trust and debugging.

The central question is whether interpretability costs performance: whether an interpretable SINDy-MPC is Pareto-competitive with black-box alternatives, or pays a measurable "price of interpretability". A fair answer is impeded by three systematic flaws in the comparison procedure itself, common in the greenhouse-control and learned-control literatures. Learned controllers are often compared against an under-tuned rule-based baseline, whereas a well-tuned agronomic heuristic is a strong competitor; setpoint-tracking error is reported in place of economic profit, although the two are not monotonically related; and, least obviously, the surrogate's hyper-parameters are chosen by open-loop prediction accuracy, on the tacit assumption that a more accurate model yields better control. This last assumption is exactly what identification-for-control has warned against for three decades [7, 8], and what objective mismatch has re-established for learned dynamics [9, 10].

We address all three by running a pre-registered, in-silico economic benchmark. The primary metric is the simulator's own economic performance indicator (EPI)—seasonal profit read directly from the reward model, not a proxy—and, because EPI does not penalise constraint violations, the primary comparison is a two-axis Pareto criterion (EPI versus violations). Baselines include a tuned rule-based heuristic and fairly implemented reinforcement-learning controllers under an equal hyper-parameter budget. The identification recipe is frozen by open-loop and transparency criteria *before* any closed-loop evaluation, following pre-registration practice [11], so that recipe selection cannot contaminate the honest comparison. We evaluate ten controllers over 20 seeds with paired non-parametric statistics [12, 13, 14].

The pre-registered hypothesis—that SINDy-MPC is non-dominated at a low price of interpretability—is not confirmed. The value of the benchmark, however, lies not in the negative result but in the mechanism it exposes: comparison with additional controllers isolates the cause of the pre-registered interpretable controller's underperformance, and this cause is specific and transferable.

The main methodological result is an identified mechanism. The STLSQ sparsity threshold removes the control-critical boiler actuator term from the model, and closed-loop EPI follows the presence of that term, whereas the open-loop prediction error is almost insensitive to its disappearance. As a consequence, a recipe-selection procedure frozen by open-loop criteria to guard against circular inference systematically certifies a control-catastrophic model, which is the pre-registration trap. These findings rest on an honest economic benchmark on a validated agronomic simulator, in which EPI is verified term-for-term against the simulator's reward, the rule-based baseline is tuned, the hyper-parameter budget is equal across methods, and the conclusions are supported by paired statistics over twenty seeds—something earlier RL-only greenhouse comparisons lacked [15]. Finally, the study reconsiders the value of interpretability itself: it lies in a model's transparency, in safety, and in the ability to flag out-of-distribution operation, rather than in economic supremacy, and parsimony of description should not be conflated with interpretability.

2. Related work

Optimal and predictive control of greenhouse climate spans van Henten's optimal-control formulation of the coupled climate–crop system [1, 16], modern MPC [3, 17], and stochastic optimal control under weather uncertainty [18]. Reinforcement learning (RL) has been compared against MPC with mixed economic outcomes: RL can raise production but often at worse profitability [19, 20, 21]. The simulator used here is GreenLight [22, 23], wrapped by GreenLight-Gym [15]. That benchmark, however, is an RL *environment*: it provides a fast differentiable simulator and benchmarks RL algorithms, but does not perform an honest economic comparison of controllers against a tuned rule-based baseline, nor does it consider interpretable sparse-identification surrogates or the open-loop/closed-loop selection question. The present work occupies that gap.

SINDy recovers parsimonious governing equations by sparse regression over a candidate library [4], extends to actuated systems [5], and combines naturally with MPC in the low-data limit [6]. Robustness has been improved by ensembling [24], relaxed and constrained optimisation [25], and mature tooling [26, 27]. The sequentially-thresholded least-squares (STLSQ) core, however, is sensitive to its sparsity threshold: too large a threshold removes physically important small-coefficient terms, degrading the dynamics [28, 29], a problem that recent optimisers explicitly target [30]. Prior work treats this as an *identification* pathology and proposes better *optimisers*; here its consequence is instead diagnosed inside an economic MPC loop, with an established causal link to profit, showing that a standard open-loop selection protocol walks straight into it.

That open-loop prediction accuracy is a poor proxy for closed-loop control is a foundational message of identification-for-control: the loop, not the fit, must arbitrate model quality [7, 31, 8, 32]. The same phenomenon was sharpened for learned dynamics as objective mismatch—one-step likelihood is not correlated with control performance [9, 33]—with related remedies in value-aware and control-oriented model learning [34, 35]. The contribution here is neither the general principle nor a new remedy, but a concrete, interpretable, and causally verified instance of it: because the surrogate is a glass box, the failure is legible term-by-term (a single removed actuator) rather than diagnosed by re-weighting a black-box model. To restore the term we use dataset aggregation (DAgger) [36], an established bridge between an MPC expert and a learned model [37, 38, 39].

3. Methods

Primary metric. All controllers are scored by the simulator's own economics rather than a proxy. At each step the GreenLight reward model returns a profit $\pi_k = g_k - c_k$, where the gain $g_k = (x_k^{\text{fruit}} - x_{k-1}^{\text{fruit}})\,\frac{10^{-6}}{\text{dmfm}}\,p_{\text{fruit}}$ is the revenue from fruit dry-matter growth (converted to fresh weight through the dry-to-fresh ratio dmfm $= 0.065$ and priced at $p_{\text{fruit}} = 1.6$ EUR/kg), and the variable cost c_k sums heating, lighting and CO_2 dosing priced at 0.09 EUR/kWh, 0.30 EUR/kWh and 0.30 EUR/kg. The seasonal indicator is EPI $= \sum_k \pi_k$ [EUR/m^2]. We verified that the per-step economics used by every controller (and by the oracle, below) are identical term-for-term to the simulator's reward, so EPI is the ground-truth objective and not an approximation. Because EPI omits constraint penalties and a controller can raise it by spending more time outside the productive corridors ($CO_2 \in [300, 1600]$ ppm, $T \in [15, 34]$ °C, RH$\in [50, 85]$%), the primary comparison is a two-axis Pareto criterion—EPI versus the seasonal count of violation steps—reported with a non-dominated frontier (Fig. 1). As a secondary, dimensionally consistent severity scalar we report `scaled_pen`, the violation area per constraint normalised by the simulator's own scaling. We do not combine EPI and violations into a single "EUR-penalised EPI", since the simulator does not price violations.

Simulator, location and data. Experiments use the GreenLight tomato model [22] through the GreenLight-Gym interface [15]. The location is Rostov-on-Don (47.24°N, 39.71°E); real weather for 2018–2023 is from ERA5/ERA5-Land via Open-Meteo, with derived sky and soil terms. A season is 60 days starting 1 March, at a 15-minute control period. We use a leakage-free year split: training on 2018–2019, in-distribution test on 2020, and 2021–2023 held out as out-of-distribution (OOD). Identification data are collected by the rule-based controller augmented with a pseudo-random binary excitation signal (PRBS, amplitude 0.3) on the actuators, with the regression condition number monitored.

Controllers. We compare ten controllers spanning five families (Table 1): a tuned rule-based agronomic heuristic (the lower reference and the DAgger expert); the proposed SINDy-MPC in four variants—the pre-registered confirmatory recipe, a boiler-preserving dense recipe, and DAgger-repaired versions of each; a reduced first-principles grey-box MPC that reuses the same solver; a neural-network MPC (NN-MPC, 64×64 surrogate); two reinforcement-learning policies, PPO [40] and SAC [41], via Stable-Baselines3 [42]; and an oracle MPC that plans over the true simulator dynamics. All learned controllers receive an equal hyper-parameter budget (16 trials). RL is trained for 2×10^5 environment steps with observation and reward normalisation (VecNormalize), whose omission we found to distort the RL result badly.

Pipeline and pre-registration. The proposed method has four stages. The rule-based controller with a superimposed PRBS signal first collects data; sparse regression then identifies an interpretable discrete one-step map $x_{k+1} = \Xi\,\Theta(x_k, u_k, d_k)$ of degree 1, low enough for analytic MPC embedding; the resulting surrogate is embedded in a CasADi/do-mpc solver that optimises an EPI-aligned objective under the climate constraints on a finite horizon; and, where needed, online adaptation (DAgger or EKF-SINDy) is added with an OOD monitor and a safe fallback. The identification recipe (denoising, optimiser and library) is not fixed a priori: it is selected in an ablation (E2) and then frozen by open-loop identification metrics (multi-step rollout stability, MPC-embeddability, transparency) *before* and *independently of* any closed-loop EPI on the held-out test season; this pre-registration guards against circular inference. The frozen confirmatory recipe

Table 1: Controllers and the role of each. The set is chosen to rule out, in turn, the alternative explanations for the pre-registered interpretable method's underperformance.

Controller	Class	Alternative explanation tested	Answer from data
Rule-based	tuned heuristic	existence of a strong simple target (also DAgger expert)	yes — EPI 4.63, the bar
Grey-box MPC	reduced physics	weakness of the MPC/library itself	no — 3.79 works; the threshold is the cause
SINDy-MPC (confirmatory)	proposed, pre-reg.	the method under test	fails: 1.18 (boiler dropped)
SINDy-MPC (dense/+DAgger)	proposed, repaired	fixability without losing interpretability	yes — 3.8–5.2, still glass-box
Oracle (3 h CEM)	true-model MPC	insufficient model fidelity	no — 1.99, over-spends
NN-MPC	black-box surrogate	a black-box advantage	no — −5.09, worst
PPO / SAC	black-box RL	interpretability as the bottleneck	no — both below the heuristic

uses the `physics_no_cross` library, degree 1, and an ensemble optimiser; a closed-loop-selected exploratory recipe is recorded separately and reported only as post-hoc sensitivity. The identical frozen recipe is loaded by every downstream experiment through a single source of truth.

Gates and statistics. A model is admitted only if it passes a transparency gate (glass-box equations, a threshold fraction of passed physical sign/dimension checks, bootstrap structural stability of the active-term set) and an MPC-embeddability gate. Both gates are deliberately open-loop; their consequences are examined below. The source of variance is the re-collection of training data per seed: each seed re-collects the identification dataset and re-fits the surrogate, so per-seed models genuinely differ. We use 20 seeds for all controllers; SAC converged on only 10 of 20 and is reported on those. Pairwise comparisons use the Wilcoxon signed-rank test paired by seed [12] with Holm correction [13], reporting Cohen's d and bootstrap 95% confidence intervals [14, 43].

The oracle. The oracle is a receding-horizon cross-entropy-method (CEM) controller that plans directly over the true simulator integrator (48 sampled action sequences, a 3-hour horizon, two refinement iterations, warm-started from the rule-based action), with the same economic objective as the MPC. The gap between it and a surrogate-based MPC isolates the effect of model fidelity. It is not an upper bound on seasonal EPI: a short receding horizon greedily maximises short-horizon profit, and the sampling optimiser degrades at longer horizons, so it is treated as a fidelity control rather than a ceiling.

4. Results

The ten controllers are chosen to isolate the cause of the interpretable controller's underperformance by ruling out the obvious alternative explanations (a weak MPC, insufficient model fidelity, a black-box advantage). Verdicts on the pre-registered hypotheses are summarised in Table 2.

The E2 ablation over denoising, optimiser, library and degree (42 recipes) selected, by open-loop rollout stability and the transparency and embeddability gates, a recipe with the `physics_no_cross` library, degree 1, and an ensemble optimiser as the confirmatory recipe, frozen before any closed-loop evaluation. Multi-step rollout divergence is negligible for this recipe (diverged-trajectory fraction ≈ 0 for budgets ≥ 3 days), so it passes the open-loop bar that pre-registration relies on. Hypothesis **H2** (a stable-rollout recipe exists) is thus met; but, as shown below, open-loop stability is necessary and not sufficient.

Table 3 reports the 20-seed closed-loop EPI, violations and severity for all ten controllers, with paired significance against the rule-based baseline; Fig. 1 shows the EPI–violation Pareto plane. Only the DAgger-repaired SINDy-MPC reaches the top: 5.23 ± 2.40 EUR/m^2, statistically indistinguishable from the rule-based 4.63 ($\Delta = +0.59$, $p_{\mathrm{Holm}} = 0.26$). Every other controller is significantly worse than the tuned heuristic:

Table 2: Hypothesis scorecard (pre-registered H1–H4, 20-seed verdicts).

Hypothesis	Verdict	Evidence
Central	not confirmed	conf+DAgger ties RB (p=0.26); confirmatory sig. worse
H1 (gap-to-oracle)	not met	oracle 1.99 < RB/PPO/grey-box
H2 (rollout stability)	met, but insufficient	RMSE stable yet closed-loop collapses
H3 (transparency)	partial	beats NN-MPC; confirmatory < grey-box
H4a (adaptation)	not supported	DAgger +0.39, p=0.088; EKF worse
H4b (OOD guard)	supported	guard −50% viol, $p \approx 10^{-24}$
H4c (no exploitation)	partial	confirmatory over-violates; guard mitigates

Table 3: E3 closed-loop economic benchmark (20 seeds), by decreasing EPI. ΔEPI and p_{Holm} are the paired Wilcoxon test vs Rule-based with Holm correction. Frontier = non-dominated in (EPI ↑, violation-steps ↓).

Controller	EPI (EUR/m^2)	Viol. steps	scaled pen	T-corr %	n	ΔEPI vs RB	p_{Holm}	Front.
SINDy-MPC (conf+DAgger)	5.23 ± 2.40	4302	426	62	20	+0.59	0.26 (ns)	•
Rule-based	4.63 ± 0.00	2458	485	95	20	—	—	•
SINDy-MPC (dense)	3.81 ± 1.27	6073	1206	65	20	−0.82	0.017	
Grey-box MPC	3.79 ± 1.27	6083	1209	65	20	−0.85	0.017	
PPO	3.36 ± 1.59	1842	392	84	20	−1.28	0.001	•
SINDy-MPC (dense+DAgger)	3.23 ± 2.21	4467	624	66	20	−1.40	0.031	
Oracle (3 h CEM)	1.99 ± 0.68	2816	379	72	20	−2.65	2e-5	
SINDy-MPC (confirmatory)	1.18 ± 4.33	5357	793	50	20	−3.46	0.009	
SAC	-2.41 ± 1.24	1683	179	82	10	−7.04	0.010	•
NN-MPC	-5.09 ± 3.11	4368	767	70	20	−9.72	2e-5	

the grey-box MPC (3.79, $p_{\mathrm{Holm}} = 0.017$), PPO (3.36, $p_{\mathrm{Holm}} = 0.001$), the pre-registered confirmatory SINDy-MPC (1.18, $p_{\mathrm{Holm}} = 0.009$), SAC (−2.41) and NN-MPC (−5.09, $p_{\mathrm{Holm}} \approx 2 \times 10^{-5}$). The non-dominated frontier comprises the DAgger-repaired SINDy-MPC, the rule-based heuristic, PPO, and—as a degenerate minimum-violation corner with negative EPI—SAC. On the severity axis the DAgger SINDy-MPC accrues even a lower total violation severity than the rule-based controller (426 vs 485) despite more violation steps; both axes are therefore reported together. The central pre-registered hypothesis is thus not confirmed, and the only interpretable controller that matches the heuristic achieves this through a repair whose mechanism is examined next.

Mechanism: sparsity hides the actuator. The pre-registered confirmatory recipe is the weakest of the interpretable controllers (1.18 ± 4.33, against 4.63 for the rule-based baseline and 3.79 for the grey-box MPC that shares its solver and feature library). Yet the recipe was not selected for weak closed-loop control: it was frozen, before any closed-loop evaluation, as the most parsimonious model passing the open-loop and transparency gates. The recipe that pre-registration certified as identification-optimal is thus the recipe that fails in the loop.

We isolate the cause with a one-parameter ablation. Holding the feature library (`physics_no_cross`, degree 1) and the MPC fixed, the STLSQ sparsity threshold λ is swept over $[10^{-6}, 10^{-1}]$; at each value,

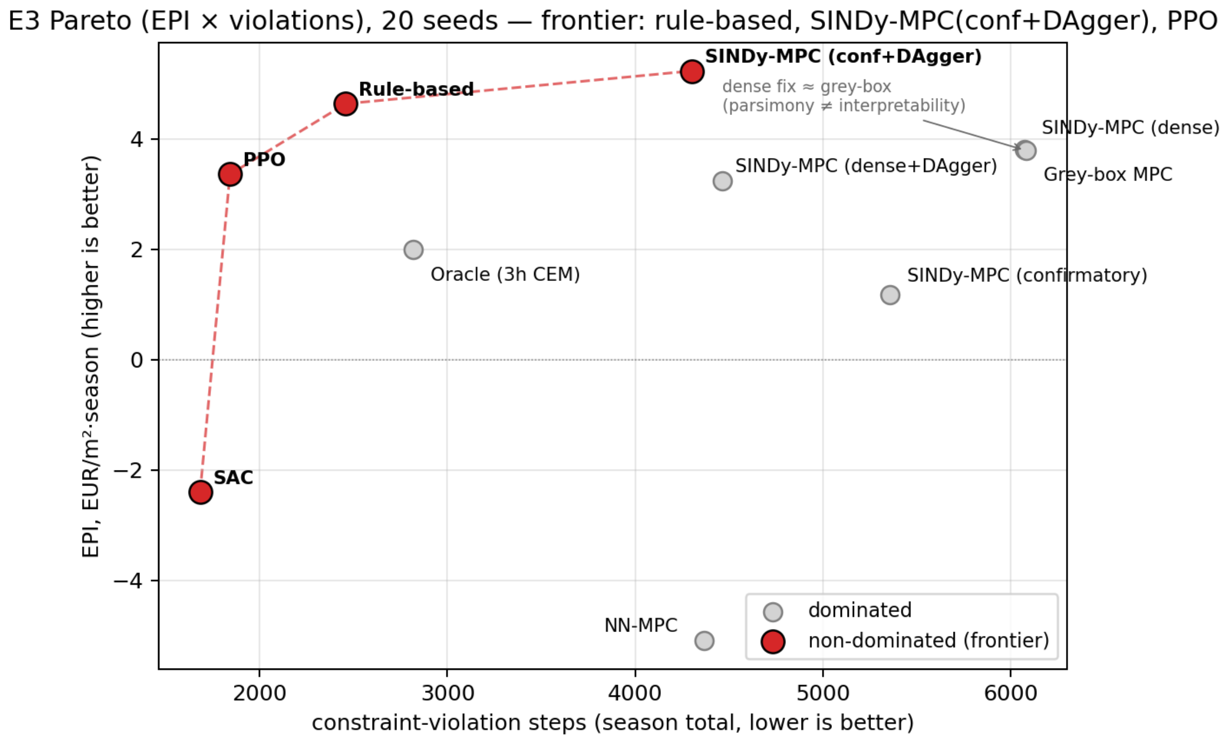

Figure 1: E3 Pareto plane (EPI × violations, 20 seeds). The non-dominated frontier (red) is rule-based, SINDy-MPC(conf+DAgger), PPO, and SAC (degenerate minimum-violation corner). The repaired dense SINDy-MPC coincides with the grey-box model: parsimony is not interpretability.

Table 4: Sparsity-threshold sweep (λ) of the `physics_no_cross` degree-1 surrogate, 20 seeds. The boiler coefficient and closed-loop EPI collapse around $\lambda\approx0.05$ while open-loop rollout-RMSE stays flat.

λ	nonzero	$\Xi_{u_{\mathrm{Boil}}\to T_{\mathrm{in}}}$	rollout-RMSE	EPI
10^{-6}	54.0	0.036	2.25	3.79
10^{-3}	49.9	0.036	2.25	3.81
0.01	41.8	0.036	2.27	4.00
0.02	38.4	0.035	2.27	4.67
0.03	34.8	0.032	2.28	3.49
0.04	30.8	0.021	2.37	1.14
0.05	27.7	0.006	2.38	0.99
0.10	20.3	0.000	2.39	2.49

over 20 seeds, we record the number of surviving coefficients, the signed boiler coefficient in the temperature equation $\Xi_{u_{\mathrm{Boil}}\to T_{\mathrm{in}}}$ (the greenhouse's only modelled active heating actuator), the *open-loop* rollout-RMSE of T_{in} on the deployment season, and the *closed-loop* EPI (Fig. 2, Table 4).

Two quantities move in opposite directions. As λ increases from 10^{-6} to 0.1 the model becomes more parsimonious (surviving terms 54 → 20) and the boiler coefficient decays and vanishes: $\Xi_{u_{\mathrm{Boil}}\to T_{\mathrm{in}}} \approx 0.035$ for $\lambda \leq 0.02$, 0.032 at $\lambda = 0.03$, 0.006 at $\lambda = 0.05$, and exactly 0 at $\lambda = 0.1$. Closed-loop EPI tracks this collapse—from a peak of 4.67 at $\lambda = 0.02$ to 0.99 at $\lambda = 0.05$, exactly where the boiler term is thresholded out. Once the surrogate contains no heating actuator, the MPC cannot plan heating, and in a heating-dominated spring the loss is severe. The open-loop metric, by contrast, is almost inert: rollout-RMSE moves only from 2.25 to 2.39 (a 6% change) while EPI moves by a factor of five and the boiler term disappears. The pre-

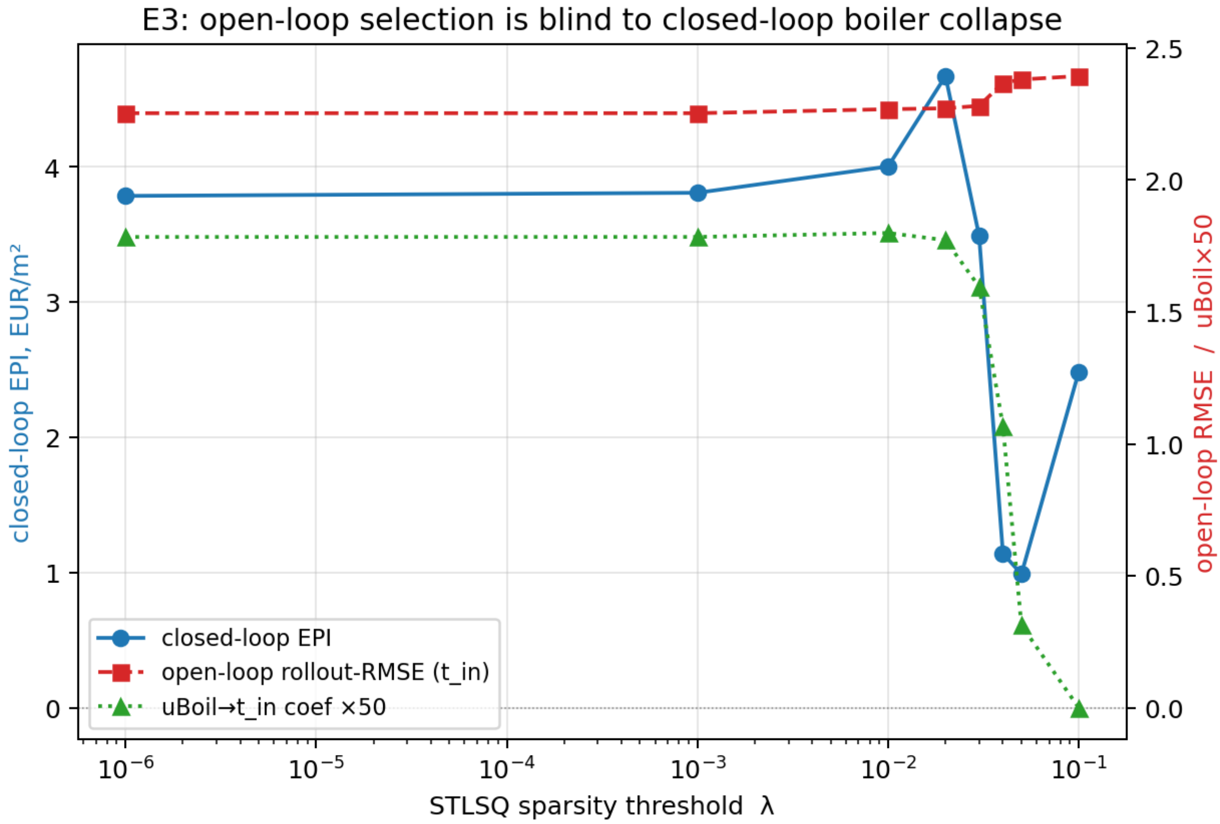

Figure 2: Sweeping the STLSQ sparsity threshold λ: closed-loop EPI (blue) collapses as the boiler coefficient $u_{\text{Boil}} \to T_{\text{in}}$ (green) is thresholded to zero, while the open-loop rollout-RMSE (red) stays flat. Open-loop model selection is insensitive to the closed-loop failure.

registered open-loop criterion is therefore insensitive to the closed-loop failure it causes. This is a concrete, physically legible instance of the mismatch between predictive accuracy and control performance [9, 10], and a sparse-identification realisation of the identification-for-control principle that the closed loop must arbitrate model quality [7, 8].

If removal of the boiler term is the cause, restoring it should recover performance, and two independent interventions do so (Table 3). A dense recipe (`stlsq`, $\lambda = 10^{-3}$) keeps $\Xi_{u_{\text{Boil}} \to T_{\text{in}}} \approx 0.034$ and lifts EPI from 1.18 to 3.81; DAgger applied to the confirmatory recipe reintroduces the term through closed-loop data and lifts EPI to 5.23, indistinguishable from the rule-based baseline. Across all SINDy-MPC variants, EPI is monotone in the presence of the boiler coefficient: whenever $\Xi_{u_{\text{Boil}} \to T_{\text{in}}} \neq 0$, it rises from ≈ 1 to 4–5 EUR/m^2. The boiler coefficient is a one-dimensional causal proxy for economic competence.

The dense fix is not a black box: it is the same glass-box map with more non-zero terms, and the repaired dense SINDy-MPC (3.81) coincides with the first-principles grey-box MPC (3.79) to within noise (Fig. 1). Once over-sparsification is removed, the interpretable surrogate collapses onto the physical model it was meant to replace. What sparsity removed was neither accuracy nor transparency but the controllability conferred by a single small-magnitude actuator term. Two practical implications follow. Sparsity should not be a model-selection objective when the task is closed-loop control [34, 28, 30]; and a transparency gate that certifies a model whose key actuator has been zeroed is incomplete—a dropped key actuator should itself be a gate failure.

Model fidelity does not explain the shortfall. If the shortfall were due to an inaccurate surrogate, an MPC over the true model should win. It does not: the oracle returns only 1.99 ± 0.68 EUR/m^2 (Table 3), below the heuristic, the DAgger and dense/grey-box models, and PPO. A horizon probe (Fig. 3) explains why: EPI falls rather than rises with horizon (+0.26 at 3 h, +0.04 at 12 h, −0.51 at 24 h on a 14-day window where the rule-based controller scores +0.63). Fruit growth and revenue rise toward the heuristic's level as the horizon lengthens, but cost, chiefly electricity and CO_2, rises faster, so the oracle attains comparable production more expensively. This is greedy over-spend from a short receding horizon, not a fidelity deficit, so the "gap-to-oracle" framing (hypothesis **H1**) is not a meaningful ceiling here. It is a structural property of short-horizon economic MPC, not an experimental error: the oracle correctly maximises the very quantity the simulator rewards, only greedily and over too short a window.

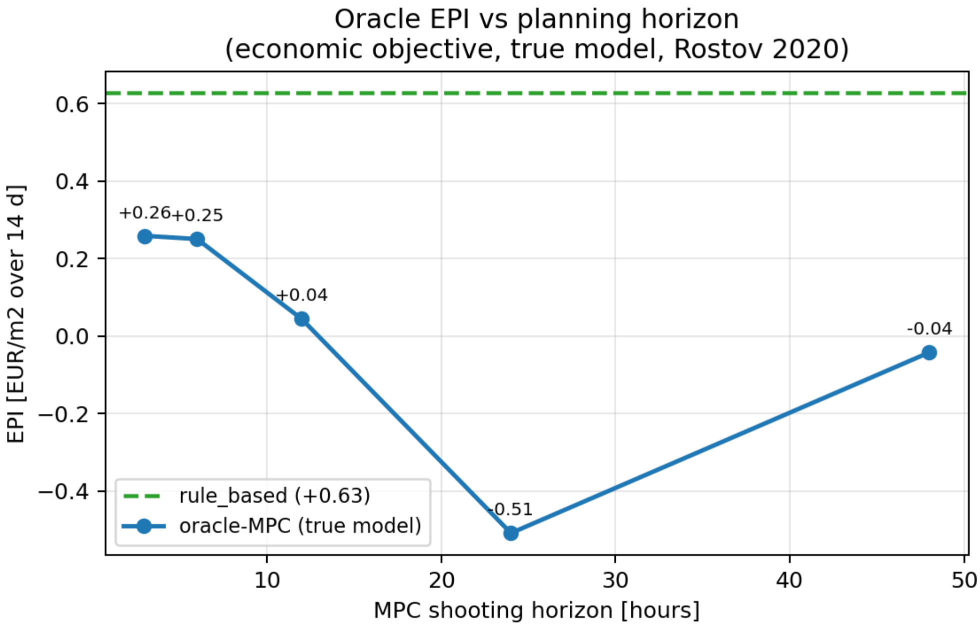

Figure 3: The oracle is not a ceiling. EPI of the true-model economic MPC falls as its planning horizon lengthens and stays below the rule-based heuristic (dashed): a short receding horizon greedily over-spends.

Interpretability is not the bottleneck. A black-box learner does not win either. RL baselines are sensitive to normalisation: a naive PPO scores -13 EUR/m^2, whereas standard observation/reward normalisation lifts it to +3.4—a reproducibility lesson rather than a property of RL [44]—yet even the normalised PPO (3.36) remains significantly below the heuristic over 20 seeds, and SAC is unstable, diverging on 10 of them. The black-box NN-MPC, in turn, is the worst controller in the study (-5.09 EUR/m^2), beaten by every interpretable SINDy variant.

Online adaptation is a null. Under spring OOD shifts (2021–2023, same planting season), the offline surrogate already generalises well—mean EPI ≈ 0, beating the rule-based controller's -2.9 under the same shift—leaving little gap for adaptation. Over 60 (shift $\times$ seed) pairs, DAgger improves on the static offline model by only +0.39 EUR/m^2 ($p = 0.088$, not significant), and online EKF-SINDy is significantly worse (-2.17, $p < 10^{-6}$); an earlier smaller sample ($n = 9$) suggesting a significant DAgger gain does not survive. Hypothesis **H4a** is not supported. The failure under shift is not gradual coefficient drift but a data problem: single-season on-shift data are intrinsically confounded ($\mathrm{corr}(u_{\mathrm{Vent}}, T_{\mathrm{in}}) = +0.67$ against +0.03 for multi-year offline data), so a re-fit mislearns the ventilation effect. What confers OOD robustness is structure (keeping the boiler term) and guarding, not coefficient adaptation.

Safety and out-of-distribution guarding. The demonstrable value of the transparent surrogate lies in safety. Its two epistemic signals—Mahalanobis distance on the exogenous inputs [45, 46] and ensemble prediction variance [47]—correlate with error and profit (Mahalanobis: +0.63 with rollout-RMSE, −0.50

with EPI; ensemble std: -0.57 with EPI), and the signal is a usable detector of high-error steps (ROC AUC $= 0.70$; Fig. 4). Gating the controller on the Mahalanobis signal, with a fallback to the rule-based policy on OOD steps, cuts seasonal violations from 1723 to 863 (a 50% reduction) while improving EPI (from -2.06 to -1.48); the reduction is highly significant over 200 pairs ($p \approx 10^{-24}$). Under injected faults (Table 5, Fig. 5), a residual-based supervisor with the same fallback significantly mitigates all six fault types, reducing violations by 1304 steps on average ($p \approx 6 \times 10^{-21}$, 120 pairs). This is the study's strongest positive result (hypothesis **H4b**).

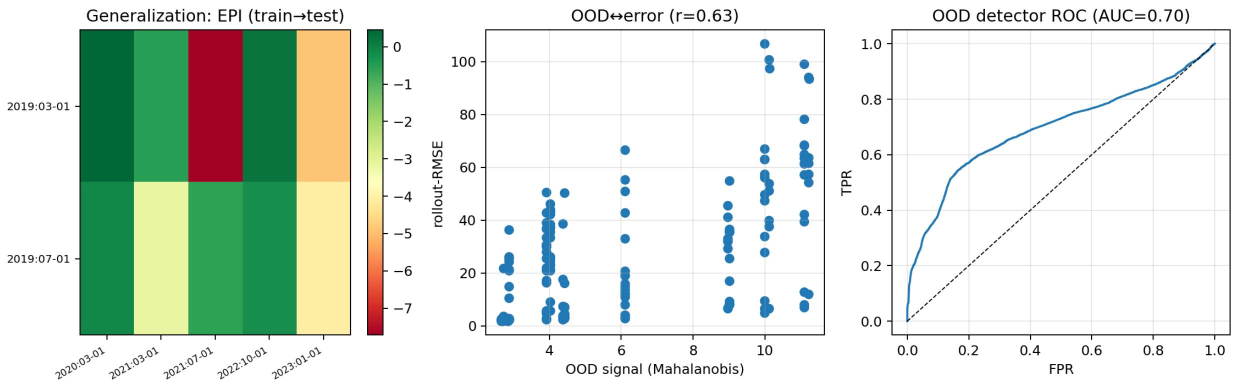

Figure 4: Generalization and OOD detection. Left: closed-loop EPI across train→test seasons (green good, red poor). Middle: the Mahalanobis OOD signal correlates with rollout error ($r = 0.63$). Right: as a detector of high-error steps the signal reaches ROC AUC $= 0.70$.

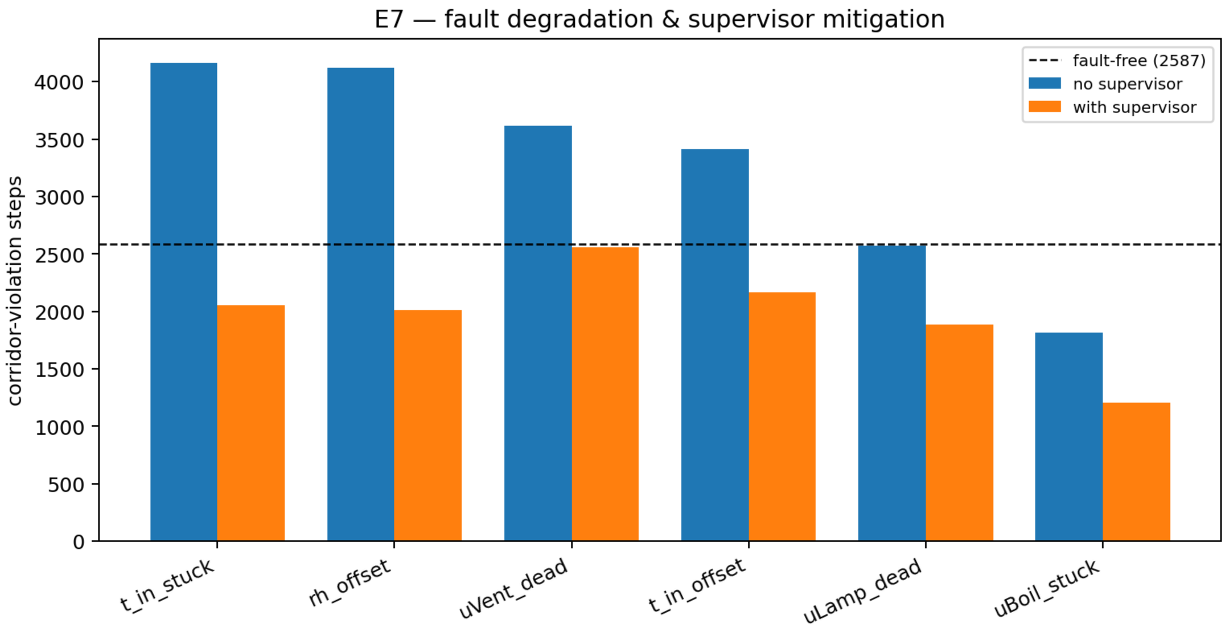

Figure 5: Fault robustness. Corridor-violation steps for six injected sensor/actuator faults, without (blue) and with (orange) the residual supervisor, against the fault-free baseline (dashed). The supervisor significantly reduces violations for every fault.

Robustness of the ranking. A sensitivity analysis (Fig. 6) shows the ranking is governed by prices more than by design choices: the EPI range from the fruit price (19.7) and energy price (13.3 EUR/m^2) dwarfs that from crop-parameter uncertainty (2.7), the sparsity threshold (1.3) and the MPC horizon (0.6).

Table 5: Fault injection: EPI and violation steps without and with the residual supervisor; fault-free baseline EPI 1.94, viol. 2587. The supervisor reduces violations for all six faults ($p \approx 6 \times 10^{-21}$ overall).

Fault	EPI (unsup)	viol (unsup)	EPI (sup)	viol (sup)
T_{in} stuck	0.86	4163	0.99	2055
RH offset	−0.07	4125	0.90	2013
u_{Vent} dead	2.01	3617	2.14	2558
T_{in} offset	1.62	3412	1.12	2165
u_{Lamp} dead	1.99	2574	2.40	1885
u_{Boil} stuck	−3.18	1816	−2.16	1208

The ranking is not robust to the energy price: when energy is expensive, the thrifty grey-box model overtakes the energy-intensive heuristic. A single-scalar ranking is therefore unreliable, and the Pareto structure is the preferred representation.

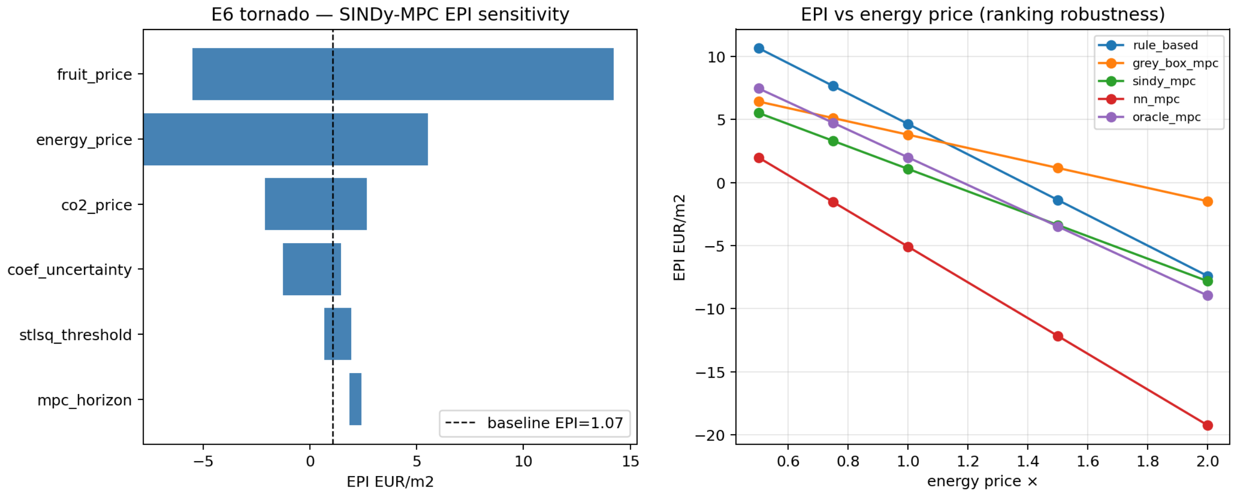

Figure 6: Sensitivity analysis. The EPI range is dominated by prices (fruit, energy) more than by design choices (crop uncertainty, sparsity threshold, horizon); the ranking is not robust to the energy price.

5. Discussion

Surrogate selection for control. The central lesson is procedural. Pre-registration froze the recipe by open-loop metrics to avoid circular inference, and thereby selected the model that fails in the loop, because that metric is insensitive to the closed-loop consequence. This is not an argument against pre-registration but against pre-registering the wrong criterion. Two corrections follow. Sparsity should not be a model-selection objective for closed-loop control: a small-coefficient term can be dynamically negligible for prediction yet control-critical; control-relevant or value-aware selection [34, 10], closed-loop-in-the-loop validation, or optimisers protecting small important terms [30, 28] are the appropriate remedies. The transparency gate, in turn, is incomplete: a gate that certifies a model whose only heating actuator has been zeroed is not verifying control-relevant transparency; a dropped key actuator, detectable against the known actuation structure, should be a gate failure.

Parsimony and interpretability. The negative result might be read as a "price of interpretability"; it is not. The repair that recovers performance (restoring the boiler term) does not sacrifice transparency: the dense model is the same glass-box map, only less parsimonious. It coincides with the grey-box model to within noise (3.81 vs 3.79): once over-sparsification is removed, the surrogate collapses onto the physical model it replaces. What sparsity destroyed was neither accuracy nor interpretability but controllability. Optimising parsimony in the name of interpretability can silently break control.

The strength of a tuned heuristic. The heuristic's strength and the oracle's over-spend share a root cause. Seasonal profit is dominated by fruit growth over days and weeks, whereas a receding economic MPC with a several-hour horizon optimises near-term profit and greedily trims heating and CO_2 that would have paid off later. A well-tuned rule set implicitly encodes the long-horizon trade-offs a short-horizon optimiser cannot see. The productive route to beating strong heuristics is therefore not a better short-horizon surrogate but a genuinely long-horizon economic formulation—a terminal value function on crop state, or a gradient-based multi-day optimiser—which we identify as the key open problem.

Generalization and limitations. The mechanism is not specific to horticulture: any pipeline that identifies a sparse surrogate, selects sparsity by prediction-oriented criteria, and embeds it in a controller is exposed to the same failure; the greenhouse merely makes it vivid, because the missing term is a named actuator whose coefficient can be watched tracking profit. The claims are in-silico and do not transfer to a physical greenhouse without separate validation. The headline E3 result uses a single test season, so deterministic controllers have zero test-weather variance and the reported variance reflects surrogate re-identification only; a multi-season economic benchmark is needed to characterise variance fully. The DAgger parity comes at an additional closed-loop data budget (three aggregation iterations of five days). The oracle is a short-horizon CEM and not a ceiling. Finally, SAC converged on only 10 of 20 seeds and is reported as a robustness observation rather than a tuned competitor.

6. Conclusions

We measured the price of interpretability for sparse-identification MPC in greenhouse climate control under a protocol designed to be honest: economic profit from the simulator as the metric, a tuned rule-based baseline, fairly normalised RL, an equal hyper-parameter budget, and a recipe frozen by pre-registration. The pre-registered claim—SINDy-MPC Pareto-competitive at a low price of interpretability—was not confirmed; the main result is the mechanism it exposed. A tuned heuristic is a strong economic reference, and the only interpretable controller that matches it does so through a repair, necessitated by a specific, transferable failure: the STLSQ threshold removes the control-critical boiler term, collapsing closed-loop profit while leaving open-loop error unchanged, so an open-loop pre-registered selection systematically certifies a control-catastrophic model. Comparison with additional controllers rules out the intuitive alternatives: the true-model oracle over-spends and is no ceiling, a normalised PPO improves but does not win, and a black-box NN-MPC is worst. The value of the transparent surrogate is real but lies elsewhere than EPI: an OOD guard halves violations while improving profit, and a fault supervisor mitigates six fault modes. Parsimony is not interpretability, and sparsity does not belong among model-selection objectives when the downstream task is closed-loop control. Future work should pursue a genuine economic ceiling—a long-horizon or terminal-value controller accounting for multi-day fruit dynamics—control-relevant recipe selection immune to the sparsity trap, and sim-to-real validation.

Reproducibility

All experiments are fully scripted and seed-controlled. We release the controller implementations, the pre-registered frozen recipe, the distributed 20-seed runners and mergers, and the analysis producing every table and figure. EPI is read from the simulator's reward model, and its per-step economics were verified term-for-term against the oracle's objective.

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